**The passage is for the following question.**

Despite the fact that parrots are well known for their long lives and complex cognition, it remains unknown whether the two traits have influenced each other. "The problem has been sourcing good quality data," says Simeon Smeele, a doctoral student at the Max Planck Institute of Animal Behavior (MPI-AB). Understanding what has driven parrot longevity is only possible by comparing living parrots. "Comparative life-history studies require large sample sizes to provide certainty, because many processes are a play at once and this creates a lot of variation," says Smeele.

To generate an adequate sample size, scientists compiled data from over 130,000 individual parrots. This database allowed the team to gain the first reliable estimates of average lifespan of 217 parrot species. The analysis revealed a diversity in life expectancy, ranging from an average of two years for the fig parrot up to an average of 30 years for the scarlet macaw.

Next, the team employed a large-scale comparative analysis to determine whether or not parrots' renowned cognitive abilities had any influence on their longevity. They examined two hypotheses: First, that having relatively larger brains enable longer lifespans. Second, that relatively larger brains take longer to grow, and therefore require longer lifespans.

These scientists then combined the data and ran models for each hypothesis. Their results provide the first support that increased brain size has enabled longer lifespans in parrots. The findings suggest that the parrots with relatively large brains had cognitive capabilities that allowed them to solve problems in the wild that could otherwise kill them, and this intelligence enabled them to live longer lives.

Max-Planck-Gesellschaft. (2022). Unravelling the mystery of parrot longevity: Bigger brains have led some species of parrot to live surprisingly long lives, new research shows. Taken on April 14, 2022 from www.sciencedaily.com/releases/2022/03/220329152826.htm.

1) The relationship between parrot longetivity and their complex brain is difficult to determine because …
   A. There is a shortage of scientist.
   B. There is insufficient time for detailed research.
   C. Scientist need to collect data from a specific parrot species.
   D. A high number of living subjects is required for the research.
   E. The research requires the right amount of data collection.

2) From the first paragraph, it can be predicted that …
   A. We will never how parrots' complex cognition affects how long they live.
   B. Comparative life-history studies will be used to study animals other than parrots.
   C. The existence of long-lived parrots with complex cognitions will remain a mystery.
   D. Collecting high-quality data will always be a challenge in comparative life-history studies.
   E. Gaining large sample sizes for comparative life-history research will become easier in the future.

3) In relation to paragraph 1, paragraph 2 discusses …
   A. The first step and the result of the study mentioned in paragraph 1.
   B. Number of samples required to qualify the data told in paragraph 1.
   C. Data analysis related to the obstacle of research written in paragraph 1.
   D. Process of collecting the data presented in paragraph 1.
   E. An action to address the issue raised in paragraph 1.

4) Which of the following statements is INCORRECT?
   A. Parrots are compared in order to determine why they live so long.
   B. Parrots have a wide range of life expectancy.
   C. Researchers came up with theories about parrots' brains and their longevity.
   D. Large-brained parrots may have longer life spans.
   E. Hundreds of thousands of parrots with different brain sizes are collected by scientist.

5) Based on the last paragraph, it can be inferred that …
   A. Parrots with larger brains outperform those with smaller brains.
   B. Parrots have intelligence that can make them live longer than other birds.
   C. Parrots' cognitive capabilities may kill them in the wild.
   D. Big-brained parrots have cognitive capabilities that help them survive in the wild.
   E. Further research on parrots' large brain and their long lives is needed to support the current finding.

**The question is based on the following text.**

Rearing insects at home as pets may sound strange and a bit nerdy, but thousands of people all over the world have already swapped their hamsters for praying mantises or stick insects. These insects, sold at fairs and pet markets, or collected in the wild and then reared by amateurs or professionals, are gaining increased popularity and fueling a largely unknown market. Not all of them are small, crawling monsters. Some are elegant, with flower-like coloration (the Orchid Mantis, *Hymenopus coronatus*), and some are funny-looking like Pokémons (the Jeweled Flower Mantis, *Creobroter wahlbergii*). Many can be safely manipulated and cuddled as they look at you with big, cute kitty-eyes (the Giant Shield Mantis, *Rhombodera basalis*).

When choosing a pet insect, "customers" consider things such as shape, size, colors, and behaviors. They might also take into account how rare a certain species is, or how easy it is to look after. Looking at these preferences, Roberto Battiston of Museo di Archeologia e Scienze Naturali G. Zannato (Italy), William di Pietro of the World Biodiversity Association (Italy) and entomologist Kris Anderson (USA) published the first overview of the mantis pet market. Understanding how this market, still mostly unregulated, is changing, may be crucial to the conservation of rare species and promoting awareness of their habitat and place in the ecosystem.

A survey among almost 200 hobbyists, enthusiasts and professional sellers in the mantis community from 28 different countries showed that the targets of this market are indeed predictable. The typical mantis breeder or enthusiast, the study found, is 19 to 30 years old and buys mantises mostly out of personal curiosity or scientific interest. Willing to spend over $30 for a single individual, most people will prefer beautiful looking species over rare ones.

Pensoft Publishers. (2022). Are people swapping their cats and goldfish for praying mantises? New research sheds light on the pet insect market and its implications on biodiversity conservation. Taken on May 20, 2022 from www.sciencedaily.com/releases/2022/05/220519081106.htm.

6) The author’s tone towards the mantis market is …
   A. Informative
   B. Subjective
   C. Enthusiastic
   D. Informal
   E. Admiring

7) Paragraph 1 implies that …
   A. Mantis rearing has become a worldwide hobby.
   B. Mantises caught by professionals are sold in the market.
   C. Some types of mantises appear to be friendly to humans.
   D. Mantises are more popular than hamsters, for they are simpler to keep.
   E. Caring for mantises is believed to keep a person happy and enjoy life more.

8) About keeping mantises as pets, the author holds the belief that …
   A. It arose as a result of the insect market’s existence.
   B. The hobby has drawn everyone’s attention.
   C. It is more valuable than keeping hamsters.
   D. The hobby is not commonly practiced.
   E. Not everyone supports the hobby.

9) Another way to restate ideas in the second paragraph of the passage is …
   A. By understanding mantis enthusiast’s preferences, experts have developed solutions to the praying mantis market’s impact on habitat conservation.
   B. Experts gace the first overview of how the mantis pet market involves enthusiasts’ preferences.
   C. By examining mantis enthusiast’s preferences, experts investigated how far the mantis pet business can implicates habitat conservation.
   D. By looking at mantis enthusiast’s preference, experts studied how mantis pet market implicates habitat conservation.
   E. Experts analysed how long it will take until the mantis pet market has an impact on their habitat conservation.

10) The passage can be summarized as …
   A. Praying mantis is the number one preferred animal over hamsters. Keeping mantises as pets have become increasingly popular, fueling an unknown market. An outline of the market for mantis has been created by certain experts.
   B. Mantises are gaining increased popularity and fueling a largely unknown market. Some experts have produced an overview of the mantis market.

C. Mantises are more popular than hamsters all over the world for several reasons, including shape, size, color, and rarity. Some experts have created a market analysis of the mantis.
D. The average mantis breeder or enthusiast is between the ages of 19 and 30 and buys mantises for personal or scientific reasons. The majority of people will favour attractive species over rare ones.
E. Mantises are becoming more popular, stimulating a largely unknown market. An overview of the mantis market has been made by certain insect enthusiast.

**The question is based on the following passage.**

A growing number of Native American households in Nevada have no access to indoor plumbing, a condition known as "plumbing poverty," according to a new study by a team from Desert Research Institute (DRI) and the Guinn Center for Policy Priorities. The study assesses trends and challenges associated with water security (reliable access to a sufficient quantity of safe, clean water) in Native American households and communities of Nevada and also found a concerning increase in the number of Safe Drinking Water Act violations during the last 15 years.

Native American communities in the Western U.S., including Nevada, are particularly vulnerable to water security challenges. This is because of factors including population growth, climate change, drought, and water rights. In rural areas, aging or absent water infrastructure creates additional challenges.

In this study, the research team used U.S. Census microdata on household plumbing characteristics to learn about the access of Native American community members to "complete plumbing facilities," including piped water (hot and cold), a flush toilet, and a bathtub or shower. They also used water quality reports from the Environmental Protection Agency to learn about drinking water sources and health violations. According to their results, during the 30-year time period from 1990–2019, an average of 0.67 percent of Native American households in Nevada lacked complete indoor plumbing—higher than the national average of 0.4 percent. Their findings show a consistent increase in the lack of access to plumbing over the last few decades, with more than 20,000 people affected in 2019.

"Previous studies have found that Native American households are more likely to lack complete indoor plumbing than other households in the U.S., and our results show a similar trend here in Nevada," said lead author Erick Bandala, Ph.D., assistant research professor of environmental science at DRI. "This can create the quality of life problems, for example, during the COVID-19 pandemic, when lack of indoor plumbing could have prevented basic health measures like hand-washing.".

Study findings also showed a significant increase in the number of Safe Drinking Water Act violations in water facilities serving Native American Communities in Nevada from 2005 to 2020. The most common health-based violations included the presence of volatile organic compounds (VOCs), presence of coliform bacteria, and presence of inorganic chemicals.

"Water accessibility, reliability, and quality are major challenges for Native American communities in Nevada and throughout the Southwest," said coauthor Maureen McCarthy, Ph.D., research professor of environmental science and director of the Native Climate project at DRI.

The study authors hope that their findings are useful to decision-makers and members of the general public who may not be aware that plumbing poverty and water quality are significant problems in Nevada.

Desert Research Institute. (2022). Growing numbers of Native American households in Nevada face plumbing poverty, water quality problems. Taken on September 21, 2022 from https://www.sciencedaily.com/releases/2022/09/220907133213.htm.

11) What is the main idea of the text?
A. Water problems for the Native American communities in Nevada.
B. Increasing number of violations to Safe Drinking Water Act in Nevada.
C. How to improve water security for Native American communities in Nevada.
D. Reasons why Native American in Nevada are vulnerable to water security problems.
E. The effects of limited quantity of indoor plumbing in Native American households in Nevada.

12) Who is the target reader of the text?
A. Desert Research Institute
B. Decision makers in Nevada
C. Native American in the U.S.
D. Native American in Nevada
E. Native American community members

13) According to paragraph 1, what condition can be called poverty plumbing?
A. When the household do not have plumbing inside the house.

B. There is reliable access to a sufficient quantity of safe and clean water.
C. There are many violations toward Safe Drinking Water Act in the household.
D. There are increasing number of households having no access to the plumbing system.
E. When the Native Americans have to go outside their house to access the plumbing system.

14) According to paragraph 2, which of the following statements is correct?
A. Population growth in Nevada makes water consumption increase.
B. Native American communities have overcome their water security.
C. Water security is a major problem for Americans living in the western U.S.
D. Climate change is the main factor causing vulnerability to water security problems.
E. Extra challenges of water problems in rural areas are caused by their limited water infrastructure.

15) "In rural areas, **aging** or absent water infrastructure creates additional challenges." (Paragraph 2)
The word *aging* in the sentence has similar meaning to …
A. Ancient B. Deteriorate C. Develop D. Mature E. Ripen

16) According to paragraph 3, the following statements are correct, EXCEPT …
A. More than thousand of people affected because of the lack of access to the plumbing system in 2019.
B. Native American households in Nevada lacking access to plumbing have increased over the last decades.
C. A flush toilet is one of the parameters to measure the access to complete plumbing facilities in households.
D. The U.S. average of households lacking complete indoor plumbing is higher than the average of Native American households in Nevada.
E. Data from the Enviromental Protection Agency is used to get information about drinking water quality, sources, ans violations in Nevada.

17) Which of these statements is correct according to paragraph 4?
A. Indoor plumbing helps to provide basic health measures in households.
B. Many Native American households in Nevada have complete indoor plumbing.
C. There is no shower, bathtub, and flush toilet inside Native American households.
D. Basic health measures could have been prevented with a complete indoor plumbing.
E. During the pandemic, Native American households in Nevada get basic health measures easily.

18) "**This** can create the quality of life problems, for example, during the COVID-19 pandemic, when lack of indoor plumbing could have prevented basic health measures like hand-washing." (Paragraph 4)
The word *this* in the sentence refers to …
A. Covid-19 pandemic
B. Native American households
C. Lack of complete indoor plumbing condition
D. The studies about the access to indoor plumbing
E. Households who have access to indoor plumbing

19) According to paragraph 5, if someone dumped inorganic chemicals into water facilities serving Native American Communities in Nevada, then it means …
A. There will be bacteries in the water
B. Violations to Safe Drinking Water Act
C. Water quality will increase significantly
D. Native American Communities will get compensation for it
E. Water facilities will not serving Native American Communities anymore.

20) What will happen if there is no access to indoor plumbing in the households?
A. Water quality will deteriorate
B. Level of sanitation will decline
C. Plumbing poverty problems will be resolved
D. There will be violations to Safe Drinking Water Act
E. Vulnerability to water security problems will decrease